

Task 3: AI Hand Gesture Recognition Report

Submitted By: G. Pavana Deepika	Task ID: Task 3 - Computer Vision
Tech Stack: Python, OpenCV, MediaPipe	Target Domain: Real-Time Gesture Classification

1. Introduction & Project Overview

Computer vision applications bridge the gap between human interaction and machine interpretation. This report details the complete engineering of a real-time AI Hand Gesture Recognition system. By leveraging standard webcam inputs, the system continuously tracks human hand skeletons and classifies specific gestures to display dynamic visual feedback on screen.

2. Project Objectives

The primary objective of this task is to build a lightweight, robust computer vision pipeline capable of recognizing four core hand gestures in real-time:

Open Hand & Fist Detects fully extended open fingers versus closed fist configurations.	Thumbs Up & Victory/Peace Recognizes directional thumb orientations and V-sign finger extensions.
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3. Technology Stack & Libraries

- Python:** Core programming language utilized for application orchestration.
- OpenCV (`cv2`):** Handles live webcam feed capture, image processing conversions, and GUI window rendering.
- MediaPipe Hands:** Google's cross-platform ML framework optimized for high-fidelity 21 3D hand landmark coordinate tracking.

4. Pipeline & Implementation Methodology

The execution workflow is structured into real-time iterative blocks:

- Frame Capture & Mirroring:** Reads sequential frames from `cv2.VideoCapture(0)` and flips horizontally for mirror visualization.
- Color Space Conversion:** Transforms input BGR frames into RGB format required by the MediaPipe pipeline.

5. Feature Extraction & Gesture Classification Logic

The system evaluates coordinate thresholds across landmark joints to classify gestures algorithmically:

Gesture Target	Landmark Heuristic Rule	Behavioral Description
Open Hand	All finger tips above PIP joints	Index, middle, ring, and pinky fingers extended upwards.
Fist	All finger tips below PIP joints	Fingers curled inward towards the palm.
Victory / Peace	Index & middle open; others closed	V-shape formed by two fingers.
Thumbs Up	Thumb tip elevated above base joints	Thumb pointed upward while other digits remain closed.

6. Step-by-Step Execution Guide

To run the application locally in your environment:

- Install dependencies: ``pip install opencv-python mediapipe``
- Place ``gesture_recognition.py`` inside the ``task-3-hand-gesture-recognition/`` folder.
- Execute via terminal: ``python gesture_recognition.py`` and press ``q`` to exit.

7. Results & Visual Verification

The application successfully renders real-time skeleton annotations and overlays categorical text (``Gesture: Open Hand``, etc.) on the live video display window.

Status: Completed & Verified. Task 3 has been fully implemented, documented, and archived successfully under the name G. Pavana Deepika.